Ahmad Najmuddin Ibrahim

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Education

Ibaraki University, Industrial Science (Robotics Engineering) — Hitachi, Japan, 2013 – 2017

- Research on assistive grouser mechanisms for wheeled rover locomotion on soft sand inclines.
- Developed prototype rovers tested on unconsolidated sandy terrain.

Ibaraki University, Intelligent Systems Engineering — Hitachi, Japan, 2011 – 2013

- Research on prototype adaptable feet sole mechanism using vacuum-jamming mechanism

Ibaraki University, Intelligent Systems Engineering — Hitachi, Japan, 2008 – 2011

Experience

Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA), Senior Lecturer (Pensyarah Universiti, DS13-A) — Pekan, Pahang, Malaysia, Dec 2024 – present, 1 year 8 months

- Teaching undergraduate courses in robotics, mechatronics, and control systems.
- Supervising PhD and Master's students in robotics and mechatronics research.
- Principal Investigator for HVAC duct monitoring robot research project.

Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA), Pensyarah Universiti (DS51-A) — Pekan, Pahang, Malaysia, Feb 2021 – Nov 2024, 3 years 10 months

- Taught core courses including Microcontroller System, Robotic Prototype Design, and Robotics System Modelling.
- Served as Head of Programme for Bachelor of Mechatronic Engineering Technology (Robotics).

Universiti Malaysia Pahang, Lecturer — Pekan, Pahang, Malaysia, May 2017 – Jan 2021, 3 years 9 months

- Developed and delivered robotics and mechatronics courses.
- Initiated research programme on assistive grouser mechanisms for wheeled rovers.

Awards

GOLD Medal — ITEX 2024 — 2024

Awarded for the project "Remotely Underwater Cleaning Robot".

SILVER Medal — ITEX 2023 — 2023

Awarded for the project "Agronetics IoT Based Smart Farming System".

Skills

Robotic Mechanical System Design

Mechanical-Electronics Systems Integration

Discrete Element Modelling (DEM)

Mobile Robotics

Interests

Terrain-Adaptive Robot Mobility

Field & Inspection Robotics

Agricultural Robotics

Embedded Systems & Control

Languages

Malay
Native speaker

English
Fluent

Japanese
Conversational

Projects

Automated Cleaning and Monitoring of HVAC Duct Using Mobile Robot 2024 – 2026
Development of a tethered mobile robot platform for automated HVAC duct inspection and cleaning, integrating visual and environmental sensors for condition assessment.
- Grant reference RDU243812, UIC240849
- Principal Investigator

Grouser Effect Contact Modelling in Cohesive Unstructured Terrains 2023 – 2025
Investigation of grouser-soil interaction mechanisms in cohesive unstructured terrains using experimental and simulation approaches.
- Grant reference RDU230135
- Principal Investigator

Development of Autonomous Mobile Robot for Optimum HVAC Duct Cleaning 2021 – 2022
Design and development of an autonomous mobile robot for HVAC duct cleaning applications.
- Grant reference RDU212402, UIC210810
- Principal Investigator

Certificates

Graduate Engineer (Manufacturing) Jan 2018

IEEE International Professional Member Jan 2018

Graduate Technologist Jan 2018